Synoptic Question 21: The Physics of Magnetic Resonance Imaging (MRI)

Reading Passage

Read the following article and then answer the questions that follow.

Magnetic Resonance Imaging (MRI) is a non-invasive medical imaging technique that provides detailed images of internal body structures, especially soft tissue. Unlike X-rays, it does not use ionising radiation.

The physics of MRI is based on a property of atomic nuclei called "spin". The human body is mostly water, so MRI focuses on the nuclei of hydrogen atoms (protons). These protons act like tiny magnets. When a patient is placed in the strong magnetic field (B_0) of the MRI scanner, these protons align with the field.

A second, oscillating magnetic field, known as a radio-frequency (RF) pulse, is then applied. If the frequency of this RF pulse matches the natural "precession" frequency of the protons (the Larmor frequency), the protons will absorb energy and "flip" into a higher energy state. This is a resonance phenomenon.

When the RF pulse is turned off, the protons relax back to their original state, re-emitting the absorbed energy as a faint radio signal. The machine's detectors pick up this signal, and a computer uses it to construct an image. The main magnetic field is created by a large coil made of a superconducting material, which must be cooled to extremely low temperatures using liquid helium.

Data:

Data for a typical MRI scanner:

Table 1: Data for MRI Analysis

Parameter	**Value**
Main magnetic field strength, B_0	3.00 T
Mass of a proton, m_p	1.67×10^{-27} kg
Charge of a proton, e	1.60×10^{-19} C
Larmor frequency per Tesla, $\gamma/(2\pi)$	42.58 MHz T^{-1}
Energy difference between spin states, ΔE	hf_L
Planck's constant, h	6.63×10^{-34} J s
Boiling point of liquid helium	4.2 K
Latent heat of vaporization of helium, L_v	2.09×10^4 J kg^{-1}
Resistance of main coil (at 293 K)	80.0 Ω
Current in main coil, I	250 A
Total volume of magnetic field, V	0.80 m^3
Permeability of free space, μ_0	$4\pi \times 10^{-7}$ T m A^{-1}

Questions

(a) The Larmor frequency (f_L) is the frequency of the RF pulse needed for resonance, given by $f_L = (\gamma/(2\pi)) \times B_0$.

  (i) Calculate the Larmor frequency required for this 3.00 T scanner.

  (ii) This is in the radio wave part of the electromagnetic spectrum. Calculate the wavelength of these radio waves.

(b) This RF pulse "flips" the proton spins between two energy states.

  (i) Calculate the energy of a single photon in this RF pulse.

  (ii) Explain why a much higher frequency, such as in the X-ray range, is not used.

(c) The main magnetic field is produced by a superconducting solenoid.

  (i) Define a **superconductor**.

(ii) If the main coil, with a current of 250 A, were made of a normal conductor at 293 K with resistance 80.0 Ω, calculate the power that would be dissipated as heat.

(iii) Explain why this result makes a superconducting coil essential.

(d) The superconducting coil must be cooled by liquid helium.

(i) A small amount of heat (a "heat leak") causes 0.50 kg of liquid helium to boil off every hour. Calculate the energy absorbed by the helium in this time.

(ii) Calculate the rate of the heat leak in watts (J s^{-1}).

(e) The magnetic field exerts forces on charged particles.

(i) A proton is moving at 2.0×10^3 m s^{-1} perpendicular to the main B_0 field. Calculate the magnetic force on it.

(ii) Calculate the radius of the circular path this force would cause.

(f) The energy E stored in a magnetic field is given by $E = \frac{B^2}{2\mu_0} \times$ Volume.

(i) Calculate the total energy stored in the 3.00 T magnetic field of the scanner.

(ii) This energy must be safely dissipated if the magnet "quenches" (loses superconductivity). If this energy were used to heat 2000 kg of water, what would be the temperature rise? (Specific heat capacity of water = 4200 J kg^{-1} K^{-1}).

Solutions

(a) (i) $f_L = (42.58 \times 10^6 \text{ Hz T}^{-1}) \times (3.00 \text{ T}) = 1.277 \times 10^8$ Hz (or 127.7 MHz).

(ii) $\lambda = \frac{c}{f_L} = \frac{3.00 \times 10^8 \text{ m s}^{-1}}{1.277 \times 10^8 \text{ Hz}} = 2.35$ m.

(b) (i) $E = hf_L = (6.63 \times 10^{-34} \text{ J s}) \times (1.277 \times 10^8 \text{ s}^{-1}) = 8.47 \times 10^{-26}$ J.

(ii) Resonance occurs when the photon energy exactly matches the energy difference ΔE between spin states. An X-ray photon has a frequency (and thus energy) that is far too high, so it would not be absorbed by the proton spins.

(c) (i) A superconductor is a material that has zero electrical resistance when cooled below a certain critical temperature.

(ii) $P = I^2R = (250)^2 \times (80.0) = 5.0 \times 10^6$ W (or 5.0 MW).

(iii) Dissipating 5.0 MW of power as heat is not feasible. It would be impossible to cool the magnet, and the energy cost would be enormous. A superconductor, with zero resistance, dissipates no power ($P = I^2R = 0$), so the current can flow indefinitely.

(d) (i) $Q = mL_v = (0.50 \text{ kg}) \times (2.09 \times 10^4 \text{ J kg}^{-1}) = 1.045 \times 10^4$ J.

(ii) $t = 1$ hour $= 3600$ s.
Rate (Power) $= \frac{\text{Energy}}{\text{Time}} = \frac{1.045 \times 10^4 \text{ J}}{3600 \text{ s}} = 2.90$ W.

(e) (i) $F = qvB = (1.60 \times 10^{-19}) \times (2.0 \times 10^3) \times (3.00) = 9.60 \times 10^{-16}$ N.

(ii) $F_c = F_{mag} \implies \frac{m_p v^2}{r} = qvB$
$r = \frac{m_p v}{qB} = \frac{(1.67 \times 10^{-27})(2.0 \times 10^3)}{(1.60 \times 10^{-19})(3.00)} = 6.96 \times 10^{-6}$ m (or 6.96 μm).

(f) (i) $E = \frac{B_0^2 V}{2\mu_0} = \frac{(3.00)^2(0.80)}{2(4\pi \times 10^{-7})} = \frac{7.2}{2.51 \times 10^{-6}} = 2.87 \times 10^6$ J.

(ii) $Q = mc\Delta T \implies \Delta T = \frac{Q}{mc}$
$\Delta T = \frac{2.87 \times 10^6}{(2000)(4200)} = 0.34$ K (or 0.34 °C).

Cambridge International AS & A Level Physics (9702)

Paper 4 Structured Questions

Variant 2 (Higher Difficulty)

Topic: Magnetic Resonance Imaging (MRI)

Instructions
Answer all parts. Show all working in calculations. For extended-response parts, use clear physics arguments and appropriate technical vocabulary.

Comprehension passage (read before answering)
Magnetic Resonance Imaging (MRI) forms images by detecting signals from hydrogen nuclei in body water. In a strong uniform field B_0, proton spins align slightly with the field, creating a small net magnetisation. An RF pulse at the Larmor frequency transfers spins between magnetic energy states. When the pulse is removed, the spins relax and emit RF signals that are detected by receiver coils.

Spatial information is obtained by adding magnetic field gradients, so that precession frequency and phase depend on position. By combining slice selection, frequency encoding, and phase encoding, the scanner reconstructs a map of tissue properties. MRI uses non-ionising radiation, but it requires large magnetic energy storage and strict engineering control, especially during quench events when stored magnetic energy is rapidly dissipated.

Some useful data are listed below and should be interpreted in context rather than used mechanically.

Additional data for this variant

- Main field strength: $B_0 = 7.0\,\mathrm{T}$
- Gyromagnetic ratio for proton: $\gamma/(2\pi) = 42.58 \times 10^6\,\mathrm{Hz/T}$
- Proton magnetic moment: $\mu_p = 1.41 \times 10^{-26}\,\mathrm{J/T}$
- Boltzmann constant: $k = 1.38 \times 10^{-23}\,\mathrm{J/K}$
- Temperature of body tissue: $T = 310\,\mathrm{K}$
- Speed of light: $c = 3.00 \times 10^8\,\mathrm{m/s}$
- Planck constant: $h = 6.63 \times 10^{-34}\,\mathrm{J\,s}$
- Permeability of free space: $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{T\,m\,A^{-1}}$

Question 1 (25 marks)

A research MRI centre is testing a 7.0 T scanner for advanced neuroimaging. In addition to numerical performance, the team must evaluate physical limits, safety implications, and image-formation trade-offs.

(a) Calculate the proton Larmor frequency at 7.0 T. Hence determine the wavelength of the RF signal in free space. [3]

(b) The spin-state separation may be modelled by

$$\Delta E = 2\mu_p B_0.$$

(i) Calculate ΔE for a proton at 7.0 T. [2]

(ii) Show that this is consistent with $\Delta E = hf$ using your value of f. [2]

(c) The net MRI signal depends on the slight population excess in the lower spin state. Using

$$\frac{N_\uparrow}{N_\downarrow} = e^{-\Delta E/(kT)},$$

estimate the fractional population difference

$$\frac{N_\downarrow - N_\uparrow}{N_\downarrow + N_\uparrow}$$

at 310 K. Comment on what this implies about intrinsic MRI signal strength. [4]

(d) **Open-ended: gradient and localisation.** Explain how magnetic field gradients allow spatial encoding in MRI. Your answer should include:

- why position-dependent field strength produces position-dependent precession frequency,
- how slice selection is achieved,
- how frequency encoding and phase encoding together locate signal sources in 2D/3D,
- one practical trade-off involving gradient strength (e.g. resolution, scan time, stimulation risk, acoustic noise).

[6]

(e) The magnetic-energy density is $u = B^2/(2\mu_0)$.

(i) Calculate the energy stored in a uniform 7.0 T field occupying 1.20 m^3. [2]

(ii) During a quench this energy is dumped in 20 s. Estimate the mean power release. [1]

(iii) State two engineering/safety reasons why controlled quench pathways are essential. [2]

(f) **Open-ended: MRI compared with CT and X-ray imaging.** Discuss two physics-based advantages and two limitations of MRI compared with ionising-radiation imaging methods. At least one point must refer to interaction mechanisms at the atomic/nuclear scale. [3]

Mark Scheme (Variant 2 only)

Total: 25 marks

(a)
- $f = (42.58 \times 10^6)(7.0) = 2.98 \times 10^8\,\text{Hz}$ (accept 2.98–2.99 $\times 10^8$) [2]
- $\lambda = c/f = 3.00 \times 10^8/2.98 \times 10^8 \approx 1.01\,\text{m}$ [1]

(b) (i) $\Delta E = 2\mu_p B_0 = 2(1.41 \times 10^{-26})(7.0) = 1.97 \times 10^{-25}\,\text{J}$ [2]

(ii) $hf = (6.63\times10^{-34})(2.98\times10^8) = 1.98\times10^{-25}\,\text{J}$ and consistent within rounding [2]

(c)
- Evaluate $x = \Delta E/(kT) \approx 1.97 \times 10^{-25}/(1.38 \times 10^{-23} \times 310) \approx 4.6 \times 10^{-5}$ [2]
- Use small-x result or full expression to obtain fractional difference $\approx x/2 \approx 2.3 \times 10^{-5}$ (accept 2.2–2.4 $\times 10^{-5}$) [1]
- Valid comment: very small thermal polarization, hence weak net magnetization; MRI needs many spins, strong B_0, and sensitive detection/averaging [1]

(d) Open-ended (max 6). Award up to 1 mark per valid developed point:

- Larmor frequency depends on local field: $f(\mathbf{r}) = (\gamma/2\pi)B(\mathbf{r})$.
- Slice selection via RF bandwidth + gradient: only spins in field/frequency window excited.
- Frequency encoding maps one coordinate to received frequency spectrum.
- Phase encoding uses brief gradient to imprint position-dependent phase shift.
- Reconstruction (Fourier-based) combines encoded data to recover spatial image.
- Trade-off discussion developed (e.g. stronger gradients improve resolution/speed but increase peripheral nerve stimulation risk, eddy currents, acoustic noise, hardware demand).

(e) (i) $u = B^2/(2\mu_0) = 49/(8\pi \times 10^{-7}) \approx 1.95 \times 10^7\,\text{J}\,\text{m}^{-3}$;
$E = uV \approx (1.95 \times 10^7)(1.20) = 2.34 \times 10^7\,\text{J}$ [2]

(ii) $P = E/t \approx 2.34 \times 10^7/20 \approx 1.17 \times 10^6\,\text{W}$ [1]

(iii) any two developed points: avoid coil/mechanical damage from rapid heating/forces; safely vent helium to prevent asphyxiation/overpressure; protect nearby equipment/patients from abrupt field collapse hazards [2]

(f) Open-ended comparison (max 3):

- Advantages (any two): non-ionising RF photons; excellent soft-tissue contrast via relaxation physics; multi-parametric functional information.
- Limitations (any two, with max 1 if list-like): expensive/slow; contraindications with implants/ferromagnetic objects; lower sensitivity for lung/cortical bone; motion sensitivity.
- At least one point must explicitly reference microscopic interaction mechanism for full marks (e.g. nuclear spin transitions vs electron ionisation).

Sample Solution (Concise Worked Answers)

(a) $f = (42.58 \times 10^6)(7.0) = 2.98 \times 10^8\,\text{Hz}$. $\quad \lambda = \dfrac{3.00 \times 10^8}{2.98 \times 10^8} = 1.01\,\text{m}$.

(b) (i) $\Delta E = 2\mu_p B_0 = 2(1.41 \times 10^{-26})(7.0) = 1.97 \times 10^{-25}\,\text{J}$.

(ii) $hf = (6.63 \times 10^{-34})(2.98 \times 10^8) = 1.98 \times 10^{-25}\,\text{J}$, agreeing with (i).

(c)

$$\frac{\Delta E}{kT} = \frac{1.97 \times 10^{-25}}{(1.38 \times 10^{-23})(310)} = 4.6 \times 10^{-5}.$$

For small x, $e^{-x} \approx 1 - x$, so

$$\frac{N_\downarrow - N_\uparrow}{N_\downarrow + N_\uparrow} \approx \frac{x}{2} = 2.3 \times 10^{-5}.$$

Hence thermal polarization is tiny, so raw MRI signal is intrinsically weak and requires strong fields and high-SNR detection.

(d) Gradients superimpose a position-dependent field on B_0, making precession frequency position-dependent. During slice selection, an RF pulse of limited bandwidth excites only spins whose local Larmor frequencies lie in that band, selecting one slice. For in-plane encoding, one gradient during readout maps position to frequency (frequency encoding), while a separate short gradient before readout imprints position-dependent phase (phase encoding). Repeating with different phase steps fills k-space; Fourier reconstruction yields image coordinates. Stronger gradients can improve resolution and reduce encoding time, but raise acoustic noise and peripheral nerve stimulation risk.

(e) (i)

$$E = \frac{B^2}{2\mu_0} V = \frac{49}{2(4\pi \times 10^{-7})}(1.20) = 2.34 \times 10^7\,\text{J}.$$

(ii)

$$P = \frac{E}{t} = \frac{2.34 \times 10^7}{20} = 1.17 \times 10^6\,\text{W}.$$

(iii) Controlled quench circuits and venting are essential to prevent catastrophic coil heating/mechanical stress and to safely remove boiled helium gas, reducing overpressure/asphyxiation hazards.

(f) MRI uses non-ionising RF interactions with nuclear spin states, reducing ionisation risk compared with X-ray/CT (electron ionisation). MRI also provides superior soft-tissue contrast via T_1/T_2/proton-density differences. Limits include higher cost, longer scan times, sensitivity to motion, and safety constraints around metallic implants and high magnetic fields.